

Lab 46 — Log Analysis with Regular Expressions

Course: CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)

Topic 08: Software Troubleshooting — Core 2, 23% of Core 2

Exam objective: Filter and parse system logs with regular expressions to isolate the events that matter during troubleshooting (Core 2 objectives 3.1 and 4.8).

Goal

Logs contain thousands of lines and perhaps three that matter. You build the regular expressions that find those three, testing each pattern in RegexLab before applying it to real log data on the Killercoda playground.

What you'll produce

A tested regular expression library for log analysis, applied to real log files with the matches verified.

Tools and equipment

RegexLab (<https://alfredang.github.io/regexgenerator/>), Killercoda Ubuntu Playground, grep

Browser tools used in this lab

- **RegexLab** — <https://alfredang.github.io/regexgenerator/>
- **Killercoda Ubuntu Playground** — <https://killercoda.com/playgrounds/scenario/ubuntu>



RegexLab

Build, test, and debug regular expressions in real time

REGULAR EXPRESSION

`/(\w+)\s(\w+)`

☒ **g** global

☐ **i** ignore case

☐ **m** multiline

☐ **s** dotall

☐ **u** unicode

☐ **y** sticky

TEST STRING

Hello World

Foo Bar

The quick brown fox jumps over the lazy dog.

Contact: alice@example.com, bob_smith@test.org

Phone numbers: 555-123-4567 and 555.987.6543

SUBSTITUTION

Replacement string (use \$1, \$2...)

RESULT

dog.

Contact: alice@example.com, bob_smith@test. numbers: 555-123- 555.987.6543

RegexLab · client-side only · use

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RegexLab — the panels and fields this lab uses.



Killercoda Ubuntu Playground — a real Linux machine

Terminal — root@controlplane

```
$ cat /etc/os-release | grep PRETTY
PRETTY_NAME="Ubuntu 22.04.3 LTS"
$ apt-get update -qq && apt-get install -y net-tools dnsutils
Reading package lists... Done
$ ip -brief addr show
lo        UNKNOWN    127.0.0.1/8
eth0      UP           10.244.0.14/24
$ dig +short www.tertiarycourses.com.sg
104.22.35.191
$ chmod 640 ~/applus/logs/system.log && ls -l ~/applus/logs/
-rw-r----- 1 root root 118 Aug 19 09:14 system.log
$ grep -c 'ERROR' ~/applus/logs/system.log
2
$ _
```

1

Open the URL and wait for the prompt. You get root on a real Ubuntu machine — nothing to install.

3

The playground RESETS when idle. If your terminal disappears, reopen it and re-run the setup steps.

Killercoda Ubuntu Playground · <https://killercoda.com/playgrounds/scenario/ubuntu>

Killercoda Ubuntu Playground — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any

machine.

- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Open RegexLab at <https://alfredang.github.io/regexgenerator/> and review the cheatsheet covering character classes, anchors, quantifiers and groups.
2. Record the core building blocks: `\d` matches a digit, `\w` a word character, `\s` whitespace, `.` any character, `^` start of line, `$` end of line and `\b` a word boundary.
3. Build a pattern matching an IPv4 address and test it against sample text: `\bd{1,3}\.d{1,3}\.d{1,3}\.d{1,3}\b`. Record the match count.
4. Build a pattern matching a timestamp in HH:MM:SS format and record the matches: `\bd{2}:\d{2}:\d{2}\b`.
5. Build a pattern matching error severity keywords using alternation and the ignore-case flag: `(ERROR|CRITICAL|FATAL|WARN)`.
6. Build a pattern matching an email address and test it against the sample text already loaded in the tool.
7. Build a pattern matching a MAC address in colon-separated form: `([0-9A-Fa-f]{2}){5}[0-9A-Fa-f]{2}`.
8. Experiment with the flags and record what each changes: `g` for all matches, `i` for case insensitivity, `m` for multiline anchors and `s` for dot matching newlines.
9. Open the Killercodea Ubuntu playground and create a realistic log file to apply your patterns to.

```
bash cat > /root/app.log << 'EOF' 2026-08-19 09:14:22 INFO Service started on
192.168.1.10 2026-08-19 09:15:03 ERROR Connection refused from 10.0.0.55 2026-08-19
09:15:44 WARN Disk usage 91% on /dev/sda1 2026-08-19 09:16:10 ERROR Auth failed for user
admin from 203.0.113.9 2026-08-19 09:17:55 INFO Backup completed 2026-08-19 09:18:31
CRITICAL Database unreachable at 10.0.0.80 EOF cat /root/app.log
```

10. Apply the severity pattern with `grep` and confirm it returns exactly the error, warning and critical lines.

```
bash grep -inE '(ERROR|CRITICAL|WARN)' /root/app.log
```

11. Apply the IP address pattern to extract every address mentioned in the log, then sort them uniquely.

```
bash grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' /root/app.log | sort -u
```

12. Combine patterns to answer a real support question — which IP addresses appear in error lines only — and record the answer.

```
bash grep -E '(ERROR|CRITICAL)' /root/app.log | grep -oE '\b([0-9]{1,3}\.){3}[0-9]
```

```
{1,3}\b' | sort -u
```

Test it — verification

Every pattern is verified in RegexLab with its match count recorded; the severity grep returns exactly four lines from the log; the IP extraction returns four unique addresses; and the combined query correctly returns only the addresses from error and critical lines.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the <code>apt-get install</code> step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against <code>labs/tools.md</code> . All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete `worksheet.md` as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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